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Database Architecture, Users, Data Models, Schemas and Instances

Inside the World of Databases: Decoding the Blueprint



Learning Outcomes

- By the end of this lecture, students will be able to:
- * Identify different database users.
- * Understand the components of a database system.
- * Define data models.
- * Differentiate between schema and instance.
- * Understand database metadata.
- * Compare different types of data models.



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Components of a Database System

- A Database System consists of all the elements required to store, manage, retrieve, and secure data.
- Major Components
 - * Database (Actual Data)
 - * DBMS Software
 - * Application Programs
 - * Database Users
 - * Hardware
 - * Operating System
 - * Network (for distributed/client-server systems)
- Definition: A Database System is the combination of the database, DBMS, users, applications, and supporting hardware/software working together.
- Example: University Management System
 - * Students use the portal.
 - * Faculty upload marks.
 - * DBMS manages data.
 - * Database stores records.
 - * Server hosts the application.



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Types of Database Users

- Different users interact with the DBMS in different ways.
- 1. Database Administrator (DBA)
 - Responsible for
 - * Security
 - * Backup
 - * Recovery
 - * Performance tuning
 - * User management
 - Example: Oracle DBA
- 2. Database Designers
 - Responsible for
 - * Database structure
 - * Tables
 - * Relationships
 - * Constraints
 - Example: Designing an ERP database.
- 3. Application Programmers
 - Develop software that interacts with the database.
 - Examples: Banking software, Hospital Management System, Online Shopping Website
- 4. End Users
 - People who use applications without knowing SQL.
 - Examples: Students, Teachers, Customers



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Responsibilities of a Database Administrator (DBA)

- **Main Responsibilities**
 - * Install DBMS
 - * Create databases
 - * Define users
 - * Assign permissions
 - * Backup and recovery
 - * Monitor performance
 - * Security management
 - * Storage management
 - * Disaster recovery
 - * Database maintenance
- **Example: A bank DBA performs daily backups and monitors transaction performance.**



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What is a Data Model?

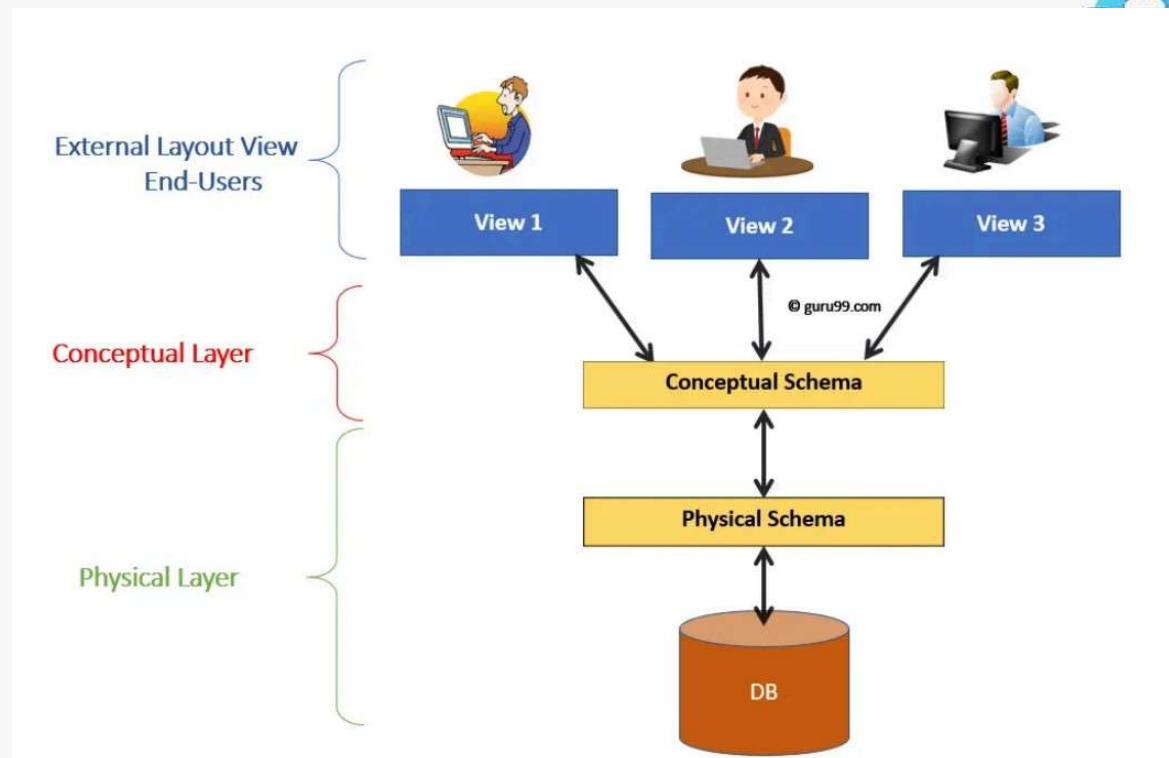
- **Definition:** A Data Model is a collection of concepts used to describe the structure of a database, the relationships among data, constraints, and operations.
- It acts as a blueprint for designing databases.
- **Why Data Models?**
 - They help in
 - * Organizing data
 - * Understanding relationships
 - * Reducing redundancy
 - * Designing efficient databases
- **Example:**
- **Student**
 - * Roll Number
 - * Name
 - * Course
- **Course**
 - * Course ID
 - * Title
- **Relationship**
- **Student enrolls in Course.**



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Types of Data Models

- 1. Conceptual Data Model
 - High-level view
 - Focuses on
 - * Entities
 - * Relationships
 - Example: ER Diagram
- 2. Logical Data Model
 - Represents
 - * Tables
 - * Attributes
 - * Keys
 - * Constraints
 - Independent of hardware.
- 3. Physical Data Model
 - Describes
 - * Storage
 - * Files
 - * Indexes
 - * Memory organization
 - Dependent on DBMS implementation.



Classification of Data Models

- Record-Based Data Models
 - * Relational Model
 - * Hierarchical Model
 - * Network Model
- Object-Based Data Models
 - * ER Model
 - * Object-Oriented Model
- Semi-Structured Data Models
 - * XML
 - * JSON
 - * Document Databases
- Example: MongoDB stores JSON-like documents.
- Note: JSON-based models are widely used in NoSQL systems for flexible data storage.



Hierarchical, Network and Relational Models

- **Hierarchical Model**
 - Structure: Tree
 - One parent → Many children
 - Example: Organization hierarchy
- **Network Model**
 - Graph structure
 - Many-to-many relationships
 - Example: Employee working in multiple projects
- **Relational Model**
 - Stores data in tables
 - Rows and Columns
 - Example: MySQL, Oracle, PostgreSQL

• Comparison

• Model	Structure	
• -----	-----	
• Hierarchical	Tree	
• Network	Graph	
• Relational	Tables	



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What is a Database Schema?

- **Definition:** A Database Schema is the logical design or blueprint of the database that defines how data is organized.
- It specifies:
 - * Tables
 - * Attributes
 - * Relationships
 - * Constraints
- A schema changes rarely.
- Example:
 - * Student (RollNo, Name, Department)
 - * Course (CourseID, Title)
 - * Enrollment (RollNo, CourseID)
- This structure is the schema.



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What is a Database Instance?

- **Definition:** A Database Instance is the actual data stored in the database at a particular moment.
- Unlike the schema, the instance changes frequently.
- **Example:**
 - Schema
 - Student (RollNo, Name)
 - Instance Today
 - | Roll | Name |
|------|-------|
| --- | ---- |
| 101 | Rahul |
| 102 | Anita |
 - Instance Tomorrow
 - | Roll | Name |
|------|-------|
| --- | ---- |
| 101 | Rahul |
| 102 | Anita |
| 103 | Riya |
- The schema remains the same, but the instance changes.



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Schema vs Instance

- | | | |
|---------------------------|-------------------------|--|
| • Schema | Instance | |
| • ----- | ----- | |
| • Blueprint | Actual Data | |
| • Rarely changes | Changes frequently | |
| • Logical design | Current contents | |
| • Created during design | Exists during operation | |
-
- Analogy
 - * House Blueprint→Schema
 - * Constructed House→Instance
 - Another Analogy
 - * Class Attendance Register
 - * Format of the register = Schema
 - * Daily attendance entries = Instance



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Metadata and Data Dictionary

- Metadata
 - * Metadata means data about data.
 - It describes
 - * Table names
 - * Column names
 - * Data types
 - * Constraints
 - * Relationships
 - Data Dictionary
 - * A repository that stores metadata.
 - Example:
 - Table
 - * Student
 - Columns
 - * RollNo INTEGER PRIMARY KEY
 - * Name VARCHAR(50)
 - * Department VARCHAR(30)
 - The above description is metadata—not student data itself.



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Real-World Example: University Database

- Schema

- Student (RollNo, Name, Department)
- Course (CourseID, Title)
- Faculty (FacultyID, Name)
- Enrollment (RollNo, CourseID)

- Instance

- Student

Roll	Name
----	-----
101	Rahul
102	Neha

- Course

ID	Course
-----	-----
CS101	DBMS
CS102	AI

- Students enroll in courses.
- Faculty teaches courses



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Summary

- Today we learned
- ✓ Components of Database Systems
- ✓ Database Users
- ✓ DBA Responsibilities
- ✓ Data Models
- ✓ Types of Data Models
- ✓ Hierarchical Model
- ✓ Network Model
- ✓ Relational Model
- ✓ Schema
- ✓ Instance
- ✓ Metadata
- ✓ Data Dictionary



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Learning Reflection

- 1. Differentiate between schema and instance with an example.
- 2. Why is the relational model more popular than hierarchical and network models?
- 3. Who is responsible for backup and security in a DBMS?
- 4. Give two examples of metadata.
- 5. Why is a data model important before implementing a database?
- 6. Identify the data model best suited for:
 - * Company organizational chart
 - * Railway reservation system
 - * Social media friendship network
 - * Student database



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Homework

- Discuss the major components of Database System and various types of database users. Define data models and briefly explain their major types, and differentiate between a database schema, database instance, and metadata. Also mention the role of a data dictionary.



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Thanks!

📌 Heads Up!

These slides are just a **guide** — not the full story!

To really learn, make sure to read your book, and explore other resources too.